

Welding reinforcement between silica glass and stainless steel using nanosecond fiber laser with chromium interlayer

Jingyu Huo^a, Jinhui Yuan^a, Qinru Chen^a, Minghuo Luo^a, Junhao Lu^a, Jiancheng Xu^a, Chuangkai Li^b, Aiping Luo^a, Jiaming Li^{a,*}, Qingmao Zhang^a

^a Guangdong Provincial Key Laboratory of Nanophotonic Functional Materials and Devices, South China Normal University, Guangzhou 510006, China

^b School of Electronic and Computer Engineering, Peking University, Shenzhen 518055, China

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ABSTRACT

Modern manufacturing industries need high strength and stable joint between glass and steel. Laser welding is an efficient technique to meet the requirement, but strength still needs to be improved. In this work, a welding reinforcement method of Cr addition was proposed to improve shear strength. The two materials were added with Cr metal film and Cr³⁺ solution in the interface, and welded by an nanosecond fiber laser. A maximum shear strength of 41.4 MPa was obtained, 40 % higher than that without Cr addition. The surface morphology and elemental distribution of the welds were observed. With Cr addition, the elemental diffusion in the interface was extended. Fe-O-Si and Cr-Si bonds were greatly increased in the welds. And microcracks were significantly reduced. In addition, the Cr layer also reduced the adverse effects caused by the different thermal expansion, making it more reliable in various industrial production fields.

1. Introduction

Stable and high-strength connection between glass and metal is one of the most indispensable technologies in the manufacturing industry of semiconductor devices, automobiles, smartphones, aircraft, high-speed rail, and medical instruments [1–6]. Traditional methods, including mechanical riveting [7], adhesive bonding [8], fusion welding [9], solid-state bonding [10], and anodic bonding [11], had been hindered by their drawbacks such as poor high-temperature resistance, low production efficiency, different thermal expansion coefficients, and low processing accuracy [12]. The high requirement of micro-joining in the manufacturing industry is hardly met in these methods.

Laser welding is a promising fusion welding method based on laser-matter interaction. Due to the advantages of small laser heat source size, high temperature, high operability during the welding process, and non-contact welding, laser welding can meet the high requirements of efficiency, strength, precision, and environmental protection in modern industrial production [12–14]. It had shown great potentials in kinds of dissimilar materials welding [15–21]. Facing the challenge of welding glass and metal, laser welding also shows great potential, and had become a research hotspot in the industry.

To precisely control the thermal effect in welding the two materials

with large differences in thermal expansion, ultrafast lasers were generally used in welding glass and metals. Richard M. Carter et al. demonstrated the feasibility of ps laser welding of several pairs of glass (molten silica, borosilicate, sapphire, BK7) and metal alloy materials (copper, aluminum, steel) [22], and investigated the influence of pulse energy and focus depth on welding strength [23]. Yasuyuki Ozeki et al. welded non-alkali glass and copper using fs laser [14]. Guodong Zhang et al. discovered a bonding effect of metal particles induced by fs laser ablation in the welding process [12]. Satoshi Matsuyoshi et al. demonstrated the feasibility of fs laser welding between glass and copper with a rough surface [6]. Yipeng Wang et al. demonstrated fs laser welding mechanism between stainless steel and soda-lime glass was based on mechanical mixing, wetting, and adsorption [13]. Lin Zhang et al. discovered the concave-convex structure and adhesive areas were the reason for successfully joining by fs laser [24]. However, these ultrafast lasers were still mainly operated in laboratory cleanrooms, instead of industrial fields, due to their complex maintenance, high cost, and high environmental requirements.

Nanosecond fiber lasers became one of the most popular light sources in laser processing industries, because of their compact structure, low cost, minimal maintenance requirements, and high stability and reliability. Akihiro Utsumi et al. demonstrated the metal particles

* Corresponding author at: South China Normal University, Guangzhou 510006, China.

E-mail address: jmli@m.scnu.edu.cn (J. Li).

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formed during ns laser welding adhered to the glass [5]. Min Zhang et al. investigated the formation mechanism of the welds under the condition of micro-arc-oxidation process for pre-treatment of the aluminum alloy surface [3]. In our previous work, optical contact between glass and stainless steel was stabilized by liquid surface tension [25]. These works had proved the feasibility of nanosecond laser welding glass and steel. However, the weak welding strength and inconvenient optical contact were still the challenges. The influence of difference in thermal expansion also needs to be solved. The post process annealing and the addition of oxide coating or activating flux were general methods to improve connection strength and reduce stress [3,26,27]. But these methods had been hindered by their drawbacks such as complex processes and long experimental period.

We have discovered that Cr is more easily diffused from steel into the glass than Fe during the welding fusion [17], which gave us an inspiration that Cr addition in the interface could probably widen elemental diffusion and reinforce the welds. In this work, Cr metal film and Cr^{3+} solution were added to the interface between silica glass and stainless steel. The microscopical mechanism of weld reinforcement was investigated. Elemental diffusion and reaction were observed in the interface of the two materials. This work aims to promote the development and application of laser welding technology for metal and glass.

2. Experimental procedure

2.1. Materials

Silica glass with the size of $\phi 15 \text{ mm} \times 1 \text{ mm}$ and 304 stainless steel with the size of $30 \text{ mm} \times 40 \text{ mm} \times 2 \text{ mm}$ were used in this work. The surface of the stainless steel was polished to a roughness of less than 30 nm. The compositions of the stainless steel and glass are shown in Tables S1 and S2, respectively. The thermal physical parameters of stainless steel and glass were shown in Fig. S1 and Table S3 [28], respectively. Before laser welding, Cr metal film of different thickness was coated on the glass substrates. The Cr^{3+} solution was added to the interface of the two materials as the intermediate liquid. Glass and steel were ultrasonically cleaned before welding and coating film.

2.2. Equipment

In this work, a nanosecond fiber laser (Strong Laser LTD, YLP-20 W, $M^2 < 1.4$) was used. The wavelength, pulse duration, and spot radius of the laser were 1064 nm, 200 ns, and 20 μm , respectively. A tensile testing machine (Shandong Ke Ce, WDW-1) was used to test the shear force of the welded sample to characterize the welding quality. The shear strength σ was obtained by Eq. (1):

$$\sigma = \frac{F}{S} \quad (1)$$

where F is shear force; S is welding area. In this work, the welding area S was fixed at $2.5 \text{ mm} \times 2.5 \text{ mm}$.

In addition, a confocal laser scanning microscope (CLSM, OLS4100) was used to measure the roughness of two substrates. The measurement area was $500 \mu\text{m} \times 500 \mu\text{m}$ and the test of each material was repeated by five times. A scanning electron microscopy (SEM, ZEISS Gemini 500) were used to observe the welds. An energy dispersive spectrometry (EDS, ZEISS Gemini 500) was used to observe the element distribution. A pipettor (DLAB Hipette 0.1–2.5 μL) was used to add the intermediate liquid between silica glass and stainless steel which formed the liquid bridge before welding. Cr metal film of different thickness was coated on the silica glass substrates using a vacuum evaporation equipment (VZZ-300). A spectrophotometer was used to evaluate the transmissivity, reflectivity and absorptivity of the silica glass coated with Cr metal film. An XPS (X-ray photoelectron spectroscopy, AXIS SUPRA) with an Al X-ray source (Al K α , 1486.6 eV) was used to characterize chemical reaction

process and products. The analysis area was $300 \mu\text{m} \times 700 \mu\text{m}$. A Raman spectroscopy with laser wavelength of 532 nm (LRS, inVia) was used to characterize the composition and distribution.

2.3. Welding systems and procedures

The welding system was shown in Fig. 1(a). The laser beam was focused on the interface between glass and stainless steel. A Galvo scanning system was used to control the scanning pattern and speed. The repetition frequency and scanning speed were fixed at 40 kHz and 60 mm/s, respectively. The experiment of each parameter was repeated by five times. The feasible laser power for welding was 1–3.5 W. The welded area scanning pattern was fixed at $2.5 \text{ mm} \times 2.5 \text{ mm}$, as shown in Fig. 1(b). It consisted of horizontal lines spaced by 20 μm to form a rectangle, which was completely filled with welds without any gaps.

When reaching the interface between glass and stainless steel, the laser was linearly absorbed by the metal through the free electrons, causing avalanche ionization, ultimately leading to the formation of a high-density plasma state. A larger molten pool was formed by the energy and significant thermal accumulation of multiple-pulsed lasers [29]. The molten materials and plasma were ejected and filled the gaps between the two materials, resulting in the formation of welds, as shown in Fig. 1(d) and (e). The shear force of the welded samples was measured by a tensile testing machine with two clampers, which were used to fasten the samples and ensure no deviation during the testing process, as shown in Fig. 1(c).

The glass and steel were pretreated before welding as the following process to ensure welding quality, as shown in Fig. 1(f):

- I The stainless steel and glass were ultrasonically cleaned in absolute alcohol for 10 min and wiped.
- II The Cr metal film (5–50 nm) was coated on the surface of the silica glass.
- III Intermediate liquid with different Cr^{3+} concentrations (0.1 μL) was added at the interface between glass and stainless steel (0.1 μL alcohol was added for optical contact in the control group) [25].
- IV Pressure (10 N) was applied to the upper surface of the glass to create an optical contact area ($\text{gap} < 200 \text{ nm}$).
- V The silica glass and stainless steel remained in optical contact due to the van der Waals forces and surface tension of the intermediate liquid after the pressure was removed.
- VI Two materials were welded by laser in the optical contact area.

2.4. Finite element model

A finite element model of laser welding of stainless steel and silica glass assisted by Cr interlayer was established using MATLAB R2020a to describe the welding process. Due to the symmetry of the laser spot, half of the welding area was modelled to balance the prediction accuracy and computation time, as shown in Fig. 2(a). Area ①, ②, ⑤, and ⑥ were silica glass, while area ③, ④, ⑦, and ⑧ were stainless steel. The yellow line between area ② and ③ was the Cr interlayer. In the coating film experiment, the Cr metal film surface hardly avoided being oxidized and formed a dense layer of Cr_2O_3 . The proportion of Cr and Cr_2O_3 in the film was calculated from the XPS data. Therefore, the film was set up as a multi-shell structure (1.86 nm Cr metal and 3.14 nm Cr_2O_3) in the finite element model, as shown in Fig. 2(c). According to the definition of optical contact ($\text{gap} < 200 \text{ nm}$), the intermediate liquid was set as a thin layer with a thickness of 200 nm. The thermal physical parameters of stainless steel and glass were shown in Fig. S1 and Table S3, respectively. To get a more reasonable combination of accuracy and computational efficiency, the finer mesh was set near the interface and coarser away from the interface of the glass and stainless steel, as shown in Fig. 2(b).

A laser scanning path was set in the simulation model. A temperature

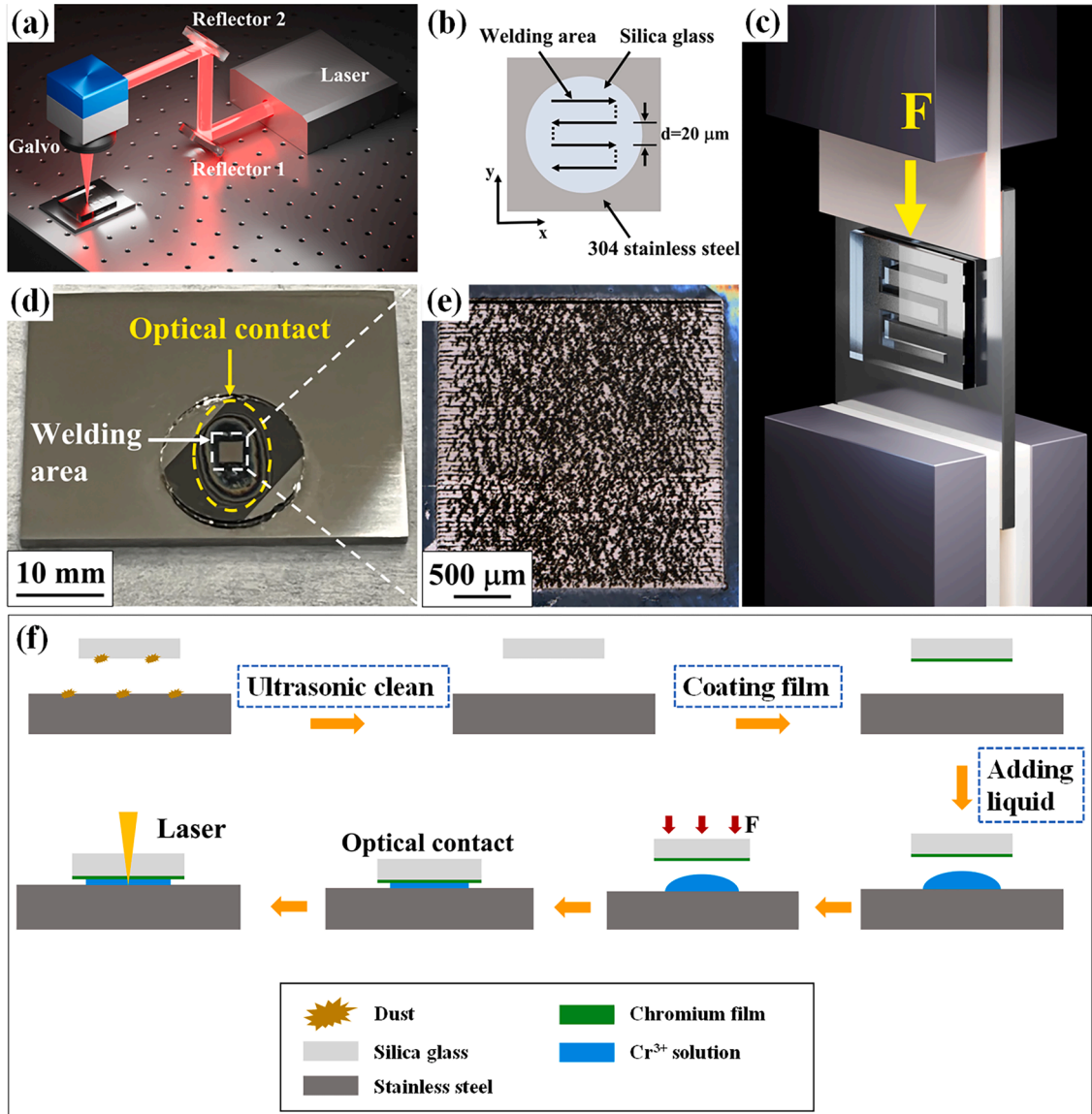


Fig. 1. (a) Diagram of laser welding system, (b) scanning pattern, (c) test method for shear strength, (d) photograph of the welded sample, (e) welding area fixed at $2.5 \text{ mm} \times 2.5 \text{ mm}$, and (f) pretreating process before welding.

probe was set on the steel surface in the middle of the path to record the average temperature within the gap between steel and glass, as shown in Fig. 2(d). Area ①, ④, ⑤, ⑥, ⑦ and ⑧ were set the infinite element to characterize the thermal diffusion process in large samples. The heat transfer process [30] was calculated by Eq. (2):

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (-k \nabla T) + Q \quad (2)$$

where Q is heat source; k is thermal conductivity; t is time; T is temperature; C_p is specific heat capacity; ρ is density.

A Gaussian heat source [31] was used to heat the substrate, calculated by Eq. (3):

$$Q = [T_g(1 - R_{ss}) + A_g] \sqrt{\frac{4 \ln 2}{\pi}} \frac{I_0}{\delta t_p} \exp \left[-\frac{r^2}{r_D^2} - \frac{z}{\delta} - 4 \ln 2 \frac{(t - t_p)^2}{t_p^2} \right] \quad (3)$$

where R_{ss} is the laser reflectivity on the stainless steel (70 %); T_g and A_g are the laser transmissivity and absorptivity of the glass with Cr metal film, respectively; I_0 is laser energy flux; r_D is the spot radius of the laser; t_p is pulse duration; δ is skin depth (10 nm) [32]; $\exp \left(-\frac{r^2}{r_D^2} \right)$ is the

Gaussian spatial distribution function; $\exp \left(-\frac{z}{\delta} \right)$ is the depth function depending attenuation of the laser pulse according to Beer-Lambert Law; $\exp \left[-4 \ln 2 \frac{(t - t_p)^2}{t_p^2} \right]$ is the Gaussian temporal distribution function.

3. Result and discussion

3.1. Effect of different cr interlayers on welding strength

The laser transmissivity, absorptivity, and reflectivity of the silica glass coated with Cr metal film were measured by the spectrophotometer, as shown in Fig. 3(a)–(c). The value of R_{ss} and T_g in Eq. (3) were set according to the measurement results.

Fig. 4(a)–(c) showed the shear strength influenced by laser powers and interlayers. Lower than 1 W, the metal surface could not form a molten pool and an effective connection with the glass; higher than 3.5 W, plenty of spatter and plasma shockwave were generated and largened the gap of the two materials, where microcracks were observed on the glass surface.

The maximum shear strength was 29.6 MPa without any Cr addition,

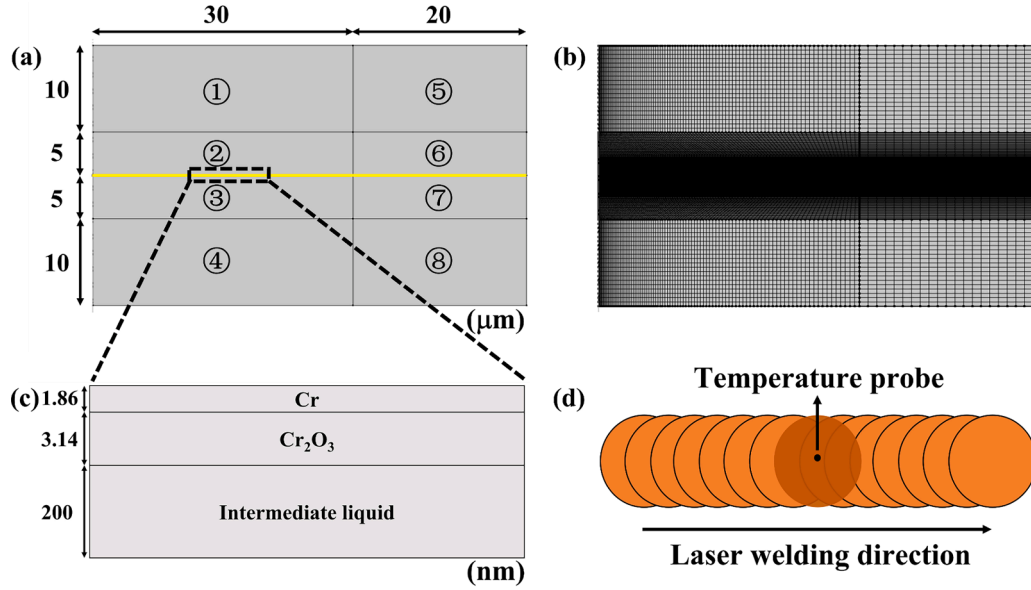


Fig. 2. (a) The geometry model, (b) the mesh, (c) the interlayer between glass and stainless steel, (d) the temperature probe.

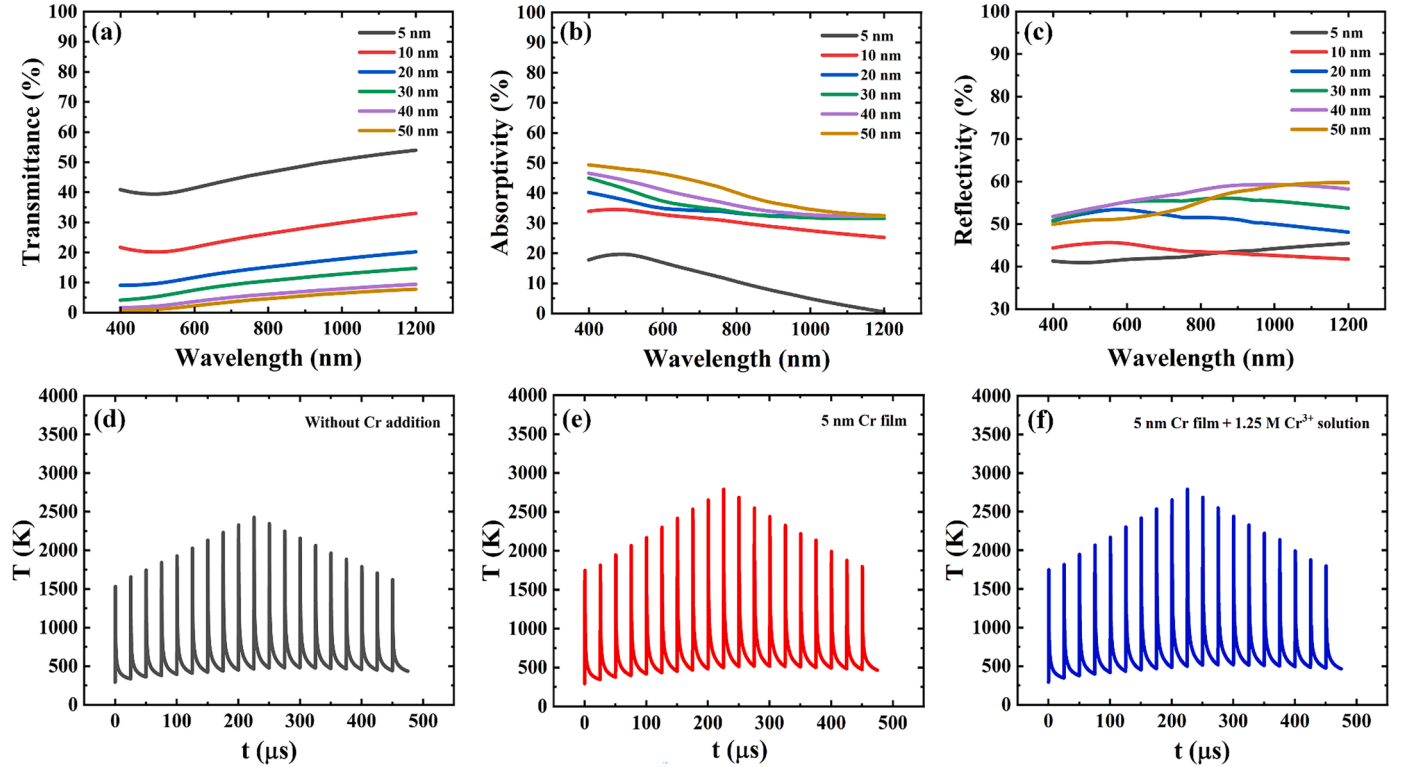


Fig. 3. (a) Transmissivity, (b) absorptivity, and (c) reflectivity spectra of silica glass coated different thickness of Cr metal film; simulation temperature on the steel surface of welding area: (d) without any Cr addition, (e) with 5 nm Cr metal film, and (f) with 5 nm Cr and 1.25 M Cr³⁺ solution.

where the weld morphology was shown in Fig. S2 (a). The strength was decreased with only Cr metal film (Fig. 4a) or Cr³⁺ intermediate liquid (Fig. 4b). Absorbed by too thick Cr metal film, excess laser energy resulted in destructive thermal stress on the glass surface. Huge amounts of cracks were observed on the edge of the welds, as shown in Fig. S3.

In contrast, combination of Cr metal film and Cr³⁺ intermediate liquid significantly increased the welding strength (Fig. 4c), where the weld morphology was shown in Fig. S2 (b). With the addition of a 5 nm Cr metal film and 1.75 M Cr³⁺ solution, the maximum shear strength of 41.4 MPa was achieved at 2.5 W, which increased by 40% than that

without any Cr addition. But the Cr³⁺ concentration exceeding 1.75 M became too viscous to form an optical contact area, and prevented the weld from forming. Fig. 3(d)–(f) represented the dependence of the simulation temperature on the different Cr interlayers. The temperature increased significantly with Cr interlayer. The detailed discussion was in Sections 3.2 and 3.3. Moreover, the surface roughness of the glass (listed in Table S4, including arithmetic mean height S_a and maximum height S_z) increased, and more molten materials filled the gap after being coated with a Cr metal film, which is another probable reason for the welding strength increasing.

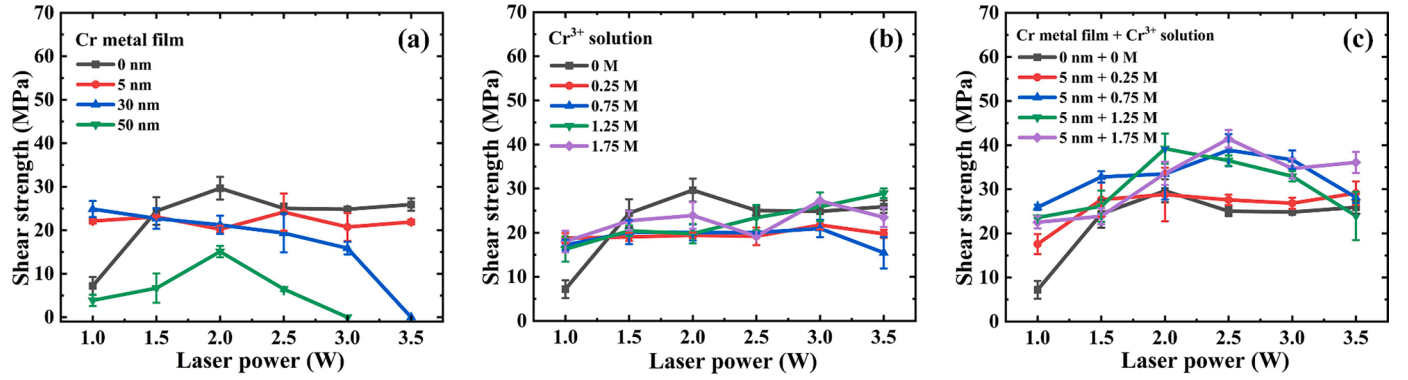


Fig. 4. Effect of different interlayers on welding strength: (a) Cr metal film, (b) intermediate liquid, and (c) Cr metal film and intermediate liquid.

3.2. Welds morphology

To investigate the effect of Cr interlayer on the welding joint, a 5 nm Cr metal film was coated on half of the glass. The laser path was scanning cross the two areas. After separating the glass and the steel, the weld morphology was observed by SEM, as shown in Fig. 5. The welds in each figure (tagged as #1, #2, #3, #4 and #5) were formed under the laser powers of 2.0, 2.5, 3.0, 3.5, and 4.0 W, respectively. There were four types of welds morphology on the material surface after separated:

- (1) No mixture of molten glass and steel was observed;
- (2) The mixture of molten glass and steel remained on the surface of the stainless steel;
- (3) The mixture of molten glass and steel remained on the surface of the glass;
- (4) The mixture of molten glass and steel was fractured and detached.

Fig. 5(a) and (c) showed the welds on stainless steel and glass surface welded with the Cr³⁺ solution (concentration of 1.25 M), respectively.

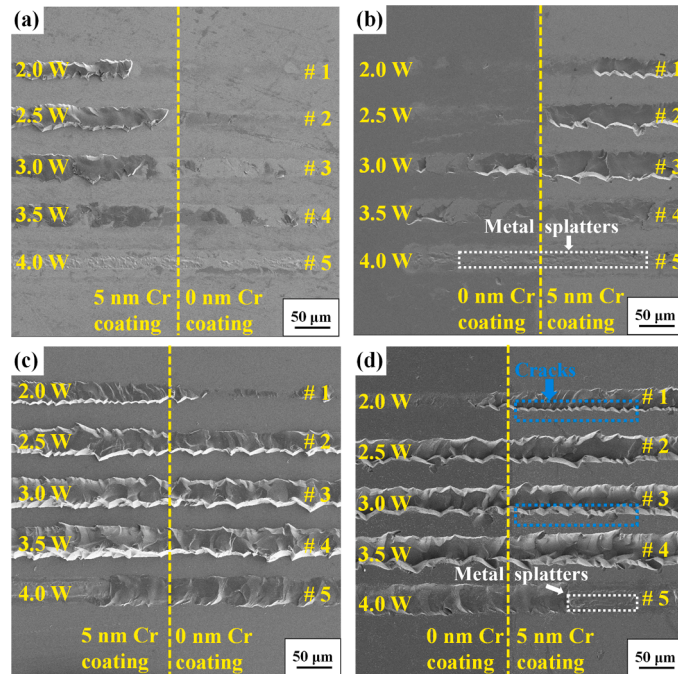


Fig. 5. Morphology of welds on stainless steel surface welded (a) with intermediate liquid, (b) without intermediate liquid; welds on silica glass surface welded (c) with intermediate liquid and (d) without intermediate liquid.

There were residual mixtures of molten glass and steel from #1 to #4. Due to excessive laser energy, more plasma and splatters were formed on the stainless steel surface at #5, providing greater recoil force and enlarging the gap. Too large gap caused failure of laser welding between the two materials. Some metal splatters remained on the glass surface and cooled down, shown as the elemental maps of welds #5 in Fig. S5 (c). Cr coating and stainless steel were melted, evaporated, and ionized to form laser-induced plasma, which was limited by the intermediate liquid and transparent glass. The subsequent laser was shielded by the plasma plume formed at the interface. The effect of excessive laser ablation on the material was reduced. The laser energy was absorbed and transferred to the surrounding materials. The concentration of thermal stress was reduced and the welding quality was improved. However, on the right-side area (without Cr metal film, only Cr³⁺ intermediate liquid) of Fig. 5(a), there were almost no remnants of molten materials on the stainless steel surface. The splatters and plasma were cooled and constrained by the intermediate liquid when the laser directly irradiated the stainless steel surface, which prevented them from forming an effective and strong connection with the glass. The thermal expansion of stainless steel was quite different from that of silica glass, so there was a risk of fracture due to the formation of large thermal stress after rapid thermal expansion and cooling shrinkage. Therefore, the buffer layer added at the interface of two materials could also reduce the adverse factors caused by the thermal expansion coefficient.

Fig. 5(b) and (d) showed the welds on stainless steel and glass surface without Cr³⁺ intermediate liquid, respectively. The right side of the welded area in Fig. 5(b) was coated with Cr metal film. The molten residue of the welded area on the right side was wider than that on the left side. The width of the welds in Fig. 5(d) was wider than that in Fig. 5(b) under the same laser power. This is because the laser energy was absorbed by the Cr metal film, resulting in a larger volume of the molten pool on the glass and more residual mixture of molten materials after being separated. The inhomogeneous distribution of thermal stress in the glass at high temperature was caused by the absence of the liquid shielding and constraint effect. A large number of cracks were observed at the edge of the welds on the glass surface, as shown in Fig. 5(d).

The diagram of connection process of four different interlayers were shown in Fig. 6. The phenomenon of splatters and plasma being cooled and constrained has been reduced by the intermediate liquid when the Cr coating was present. When the laser directly irradiated between two materials, part of the energy was absorbed by the Cr metal film and transferred to the surrounding glass. The Cr film on the lower surface of the glass and the upper surface of the stainless steel were melted simultaneously, forming two molten pools, as shown in Fig. 6(a). Optical contact was stabilized by Cr³⁺ solution surface tension. The two molten pools were mixed in a small gap, solidifying to make the welded joint more stable.

Welds on the surfaces of the stainless steel and glass were observed by SEM/EDS under different welding conditions. Fig. 7(a) and (c)

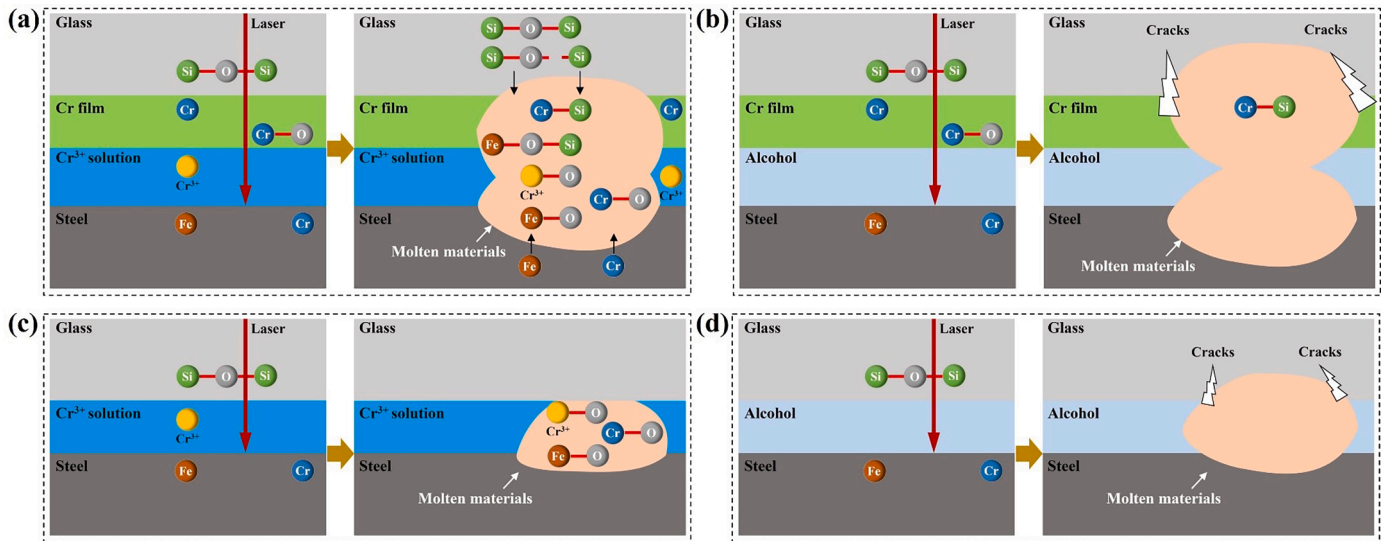


Fig. 6. Diagram of laser welding process: (a) with 5 nm Cr metal film and Cr^{3+} solution, (b) with 5 nm Cr metal film, (c) with Cr^{3+} solution, and (d) without Cr addition.

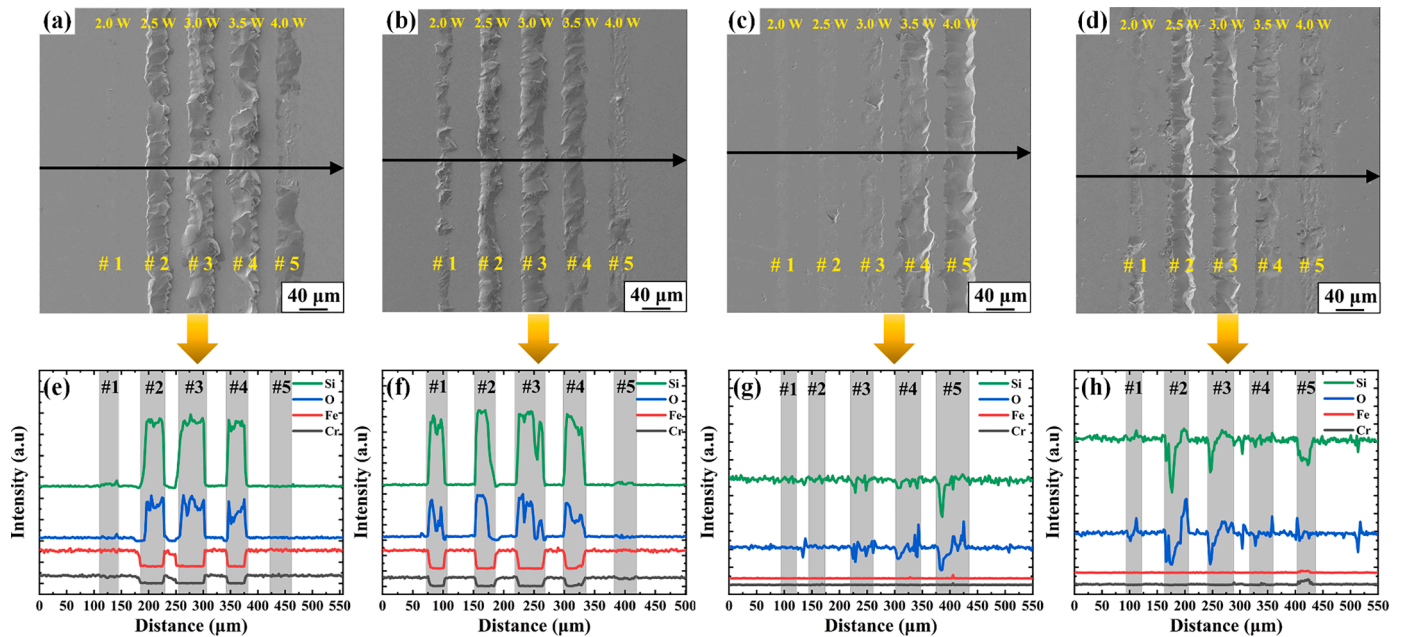


Fig. 7. Morphology of welds on steel surface: (a) without Cr addition, (b) with 5 nm Cr film and 1.25 M Cr^{3+} solution; welds on glass surface: (c) without Cr addition, (d) with 5 nm Cr film and 1.25 M Cr^{3+} solution; element distribution in the surface by EDS: (e), (f), (g) and (h) was the result of (a), (b), (c) and (d), respectively.

showed the surfaces of stainless steel and glass welded without any Cr interlayer, respectively. Fig. 7(b) and (d) showed the surfaces of stainless steel and glass welded with Cr interlayer, respectively. EDS was used to characterize the elemental distribution, as shown in Fig. 7(e)–(h), respectively. The acceleration voltage of the electron beam was 12 kV and the scanning step was 0.38 mm. The residual mixture of molten materials was observed in #1 of Fig. 7(b) under the laser power of 2.0 W compared with Fig. 7(a). The welds in Fig. 7(b) were significantly narrower and smoother than those in Fig. 7(a), as shown in Fig. 7(e) and (f). This indicated that the laser energy was absorbed by the Cr interlayer and transmitted to surrounding materials, leading to an increase of welding temperature. Therefore, the mixture of materials with Cr interlayer was formed with lower laser power (same temperature without Cr interlayer), preventing cracks produced on the glass from inhomogeneous stress.

The cross-section with different interlayers was observed by using

SEM/EDS. As shown in Fig. 8, the interface between glass and steel was smooth without any obvious mechanical interlocking structure. EDS was used to scan along the arrows in Fig. 8(a)–(c), and (d) to characterize the elemental distribution of the cross-section, and the results were shown in Fig. 8(e)–(h), respectively. The cross-section of the welds with Cr metal film and Cr^{3+} intermediate liquid was shown in Fig. 8(a), and its elemental diffusion distance was 12.4 μm . Meanwhile, the elemental diffusion distance of the cross-section of the welds welded with only Cr metal film, only Cr^{3+} intermediate liquid, and no Cr interlayer were 6.2 μm , 6.9 μm , and 6.9 μm , respectively. The elemental diffusion distance of the welds increased after simultaneously adding Cr metal film and Cr^{3+} intermediate liquid. The results agreed with the welding strength difference in Fig. 4, demonstrating broadening elemental diffusion by adding Cr interlayer (Cr film and Cr^{3+} solution) is a key factor for improving welding strength.

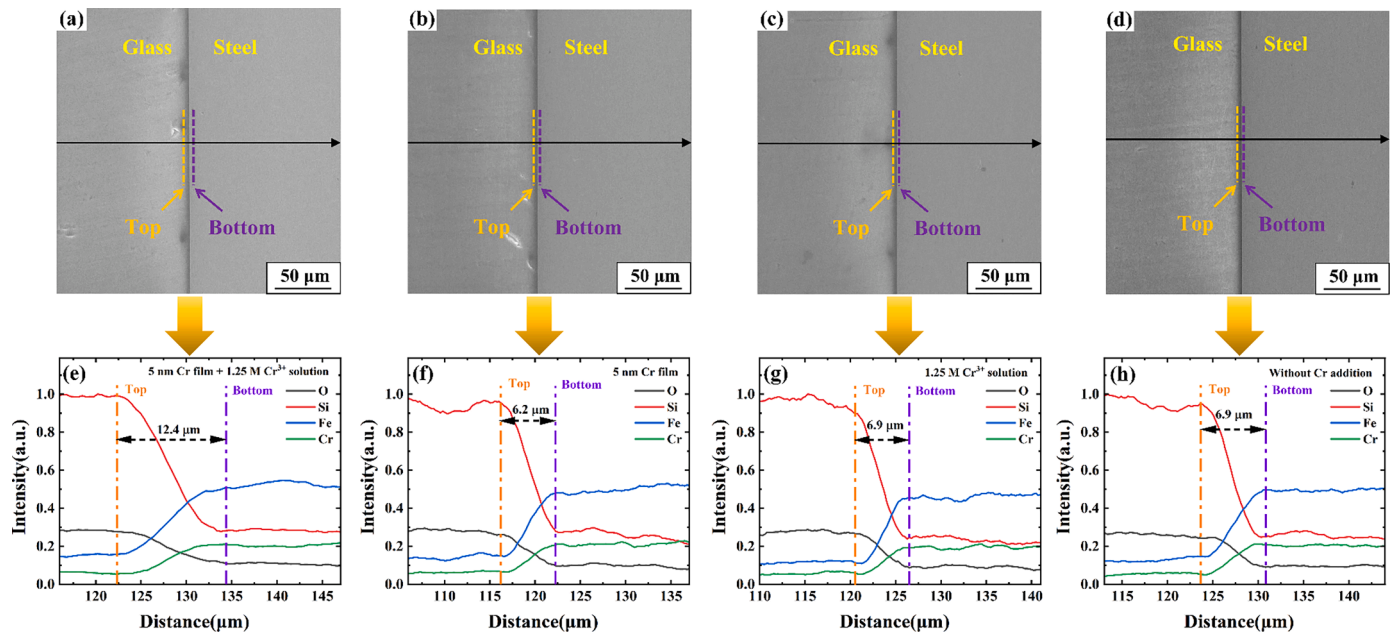


Fig. 8. Cross section of the welds: (a) with 5 nm Cr metal film and 1.25 M Cr^{3+} solution, (b) with 5 nm Cr metal film, (c) with 1.25 M Cr^{3+} solution, (d) without Cr addition; elemental diffusion by EDS: (e), (f), (g) and (h) was the result of (a), (b), (c) and (d), respectively.

3.3. Microstructure and chemical reaction

The microstructure in welds was characterized by high-resolution XPS, shown in Fig. 9. All the spectral peaks were corrected to the standard carbon C 1 s binding energy (BE) of 284.8 eV, to eliminate any other perturbation (surface state, charge effects, etc.). A shift would have occurred in the binding energy of metals in a single compound phase such as metal oxides if it became part of an inorganic complex with different phases [33].

The core levels (O 1 s, Cr 2p, Fe 2p, and Si 2p) in the substrate and welds with 5 nm Cr metal film and 1.25 M Cr^{3+} solution were analysed in Fig. 9. Binding energies of these elements in the samples were shown in Table S5. The O 1 s spectra of the steel substrate was deconvoluted into two peaks (529.1 and 530.8 eV), attributed to the oxygen in iron oxides (Fe_3O_4 and Fe_2O_3) and Cr_2O_3 [34], respectively, as shown in Fig. 9(a). The O 1 s spectra of the glass substrate coated with Cr was deconvoluted into three peaks (529.7, 532.3, and 534.3 eV), attributed to the oxygen in chromium oxides, Al_2O_3 , and SiO_2 [35], respectively, as shown in Fig. 9(b). After a Cr metal film was deposited on the glass, a Cr oxide passivation layer (Cr_2O_3) was quickly formed in the surface as it was exposed to the air. The O 1 s spectra of the welds was deconvoluted into three peaks (528.5, 529.6, and 532.0 eV), attributed to the oxygen in Cr_2O_3 , Fe_2O_3 , and SiO_2 [36], respectively, as shown in Fig. 9(c). The following analysis and fitting of Si, Fe and Cr in these areas could provide additional evidence for these products.

The Cr and Cr_2O_3 peaks were observed in the steel substrate and glass coated with Cr metal film, as shown in Fig. 9(d) and (e). After laser welding, four species of Cr components were observed in the welds. The peaks at 573.4, 574.8, 576.1, and 577.5 eV were attributed to Cr, CrSi , Cr_2O_3 , and $\text{Cr}(\text{OH})_3$ [37–39], respectively, as shown in Fig. 9(f). Interacting with laser pulse, Si-O and metallic bonds were broken, and Cr-Si bonds were formed. $\text{Cr}(\text{OH})_3$ was related to the composition of the intermediate liquid prepared.

The existence of Fe in various forms was complex, as shown in Fig. 9 (g)–(i). The peaks of the steel substrate were assigned to metallic Fe, Fe_2O_3 , Fe_3O_4 , and free Fe^{3+} [33,40–43], respectively. The peaks of welds on the steel were assigned to metallic Fe, FeO, Fe mixed in SiO_2 and Al_2O_3 , free Fe^{3+} , and the mixture of Fe^{3+} and Fe^{2+} , respectively. The mixture of welds on the steel was inhomogeneous after shear separation.

Parts of the steel substrates were exposed to air, where metallic Fe was observed. The peaks of welds on the glass were assigned to FeO, Fe mixed in SiO_2 and Al_2O_3 , free Fe^{3+} , and the mixture of Fe^{3+} and Fe^{2+} , respectively.

The signal of Si in the steel was not observed, as shown in Fig. 9(j). The peaks of the glass substrate were attributed to Si_2O_3 and SiO_2 [44], respectively, as shown in Fig. 9(k). The peaks of the welds were attributed to SiO_2 and Fe_2SiO_4 compound (Fe-O-Si bonds) [45], respectively, as shown in Fig. 9(l).

Absorbing laser energy, the Si-O bonds in the silica glass was unstable and broken. The active oxygen was generated and its coordination was unsaturated. It was connected with Fe^{2+} ions to form the Fe-O-Si bonds. The oxidation process of Fe atoms under heating conditions was reduced by the addition of Cr or Si atoms in steel. The sublimation rate of chromium oxide was faster than that of silicon oxide at high temperatures. The diffusion of the oxygen into the deeper areas of the steel was prevented by the surface passivation layer composed of Si and Cr oxides. The presence of the Fe_2SiO_4 compound was related to the Cr atoms in the passivation layer (chromium segregation on the surface area) [45].

The morphology and Fe-O-Si mapping of welds after separating the two materials were also observed by SEM and Raman spectrometer, as shown in Fig. 10. The Raman mapping was based on 1000 cm^{-1} peak, attributed to Fe-O-Si tetrahedrons [46,47]. In the welds with Cr addition shown in Fig. 10(c) and (f), Fe-O-Si bond was observed in the mixture of molten glass and steel, while no signal on the substrates. The welds were spalled on the left area in Fig. 10(b) and the steel substrate was exposed to the air, where the Fe-O-Si bond was also detected. However, Fe-O-Si bond was hardly observed on the welds without Cr addition, as shown in Fig. 10(i) and (l). It was one of the evidence that elemental diffusion was broadened by Cr addition and the welding strength was reinforced.

4. Conclusion

In this work, a welding reinforcement method was proposed for silica glass and stainless steel by a nanosecond fiber laser assisted with chromium interlayer. Cr metal film and Cr^{3+} solution were added between the two materials. Welding process parameters and bonding microscopical mechanisms were investigated. Laser power of 1–3.5 W were

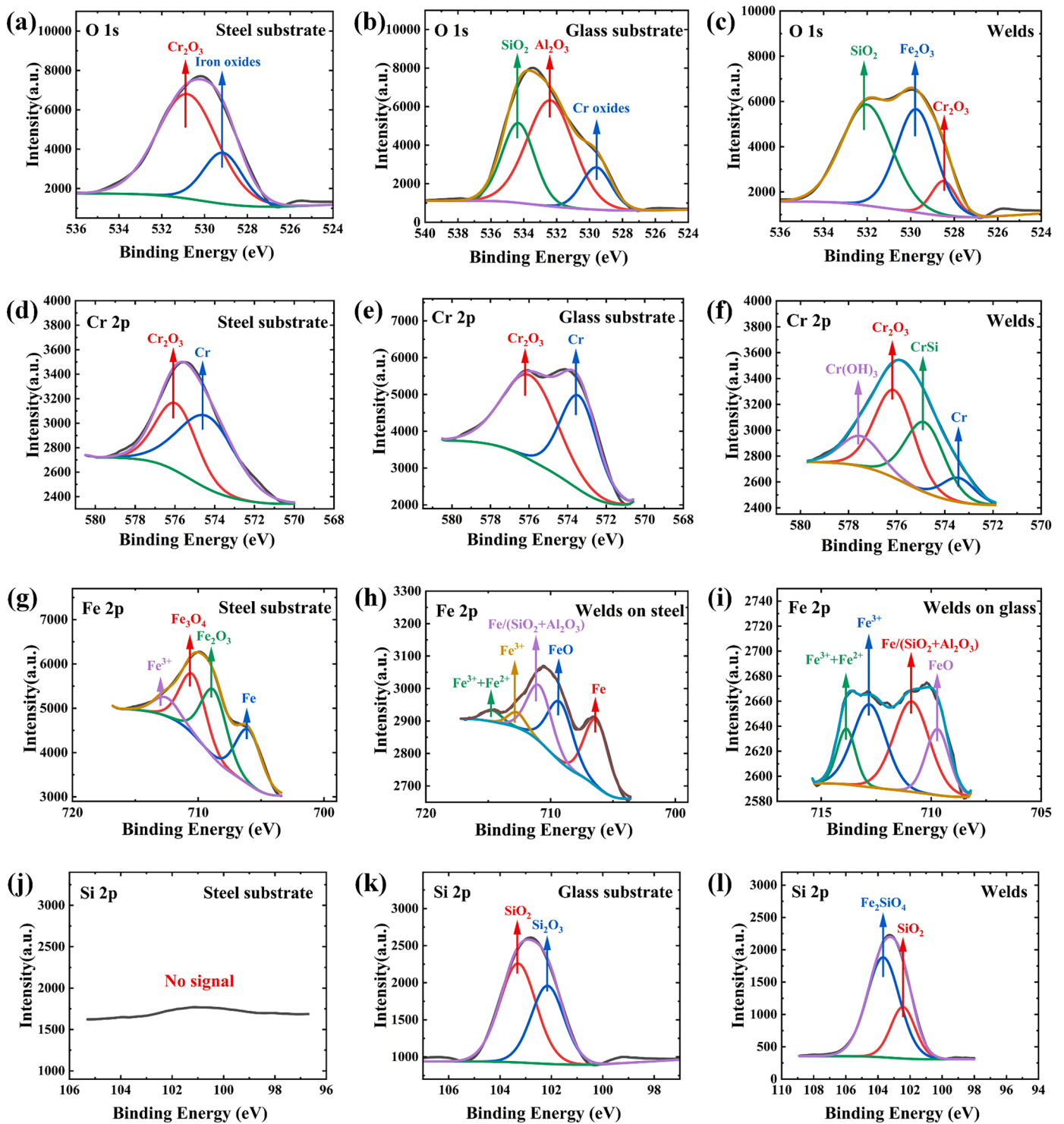


Fig. 9. High-resolution XPS spectra of O of the (a) steel substrate, (b) glass substrate, (c) welds; spectra of Cr of (d) steel substrate, (e) glass substrate, (f) welds; spectra of Fe of (g) steel substrate, (h) welds on steel surface, (i) welds on glass surface; spectra of Si of (j) steel substrate, (k) glass substrate, and (l) welds.

suitable for welding. A maximum shear strength of 41.4 MPa was obtained at 2.0 W, which increased by 40 % compared to that without Cr addition. The welds surface was observed by using SEM and CLSM. The welds with Cr addition on the stainless steel surface were narrower and smoother compared to those without Cr addition. Laser energy was absorbed and transferred to the surrounding materials by the Cr metal film and Cr^{3+} solution. The effect of excessive laser ablation on the material was mitigated. The concentration of thermal stress and significant cracks were reduced, and the welding quality was significantly

improved. The elemental diffusion measured on the cross-section of the welds was increased from 6 μm to 12 μm . Fe-O-Si and Cr-Si bonds were also observed on the welds. The content of Fe-O-Si bond was significantly increased with the addition of Cr film and solution. These above mechanisms were the main factors for the welding reinforcement. In summary, the addition of the Cr interlayer (combination of Cr film and Cr^{3+} solution) reduced the adverse effects caused by the different thermal expansion of two materials, making the joint more reliable in various industrial production fields.

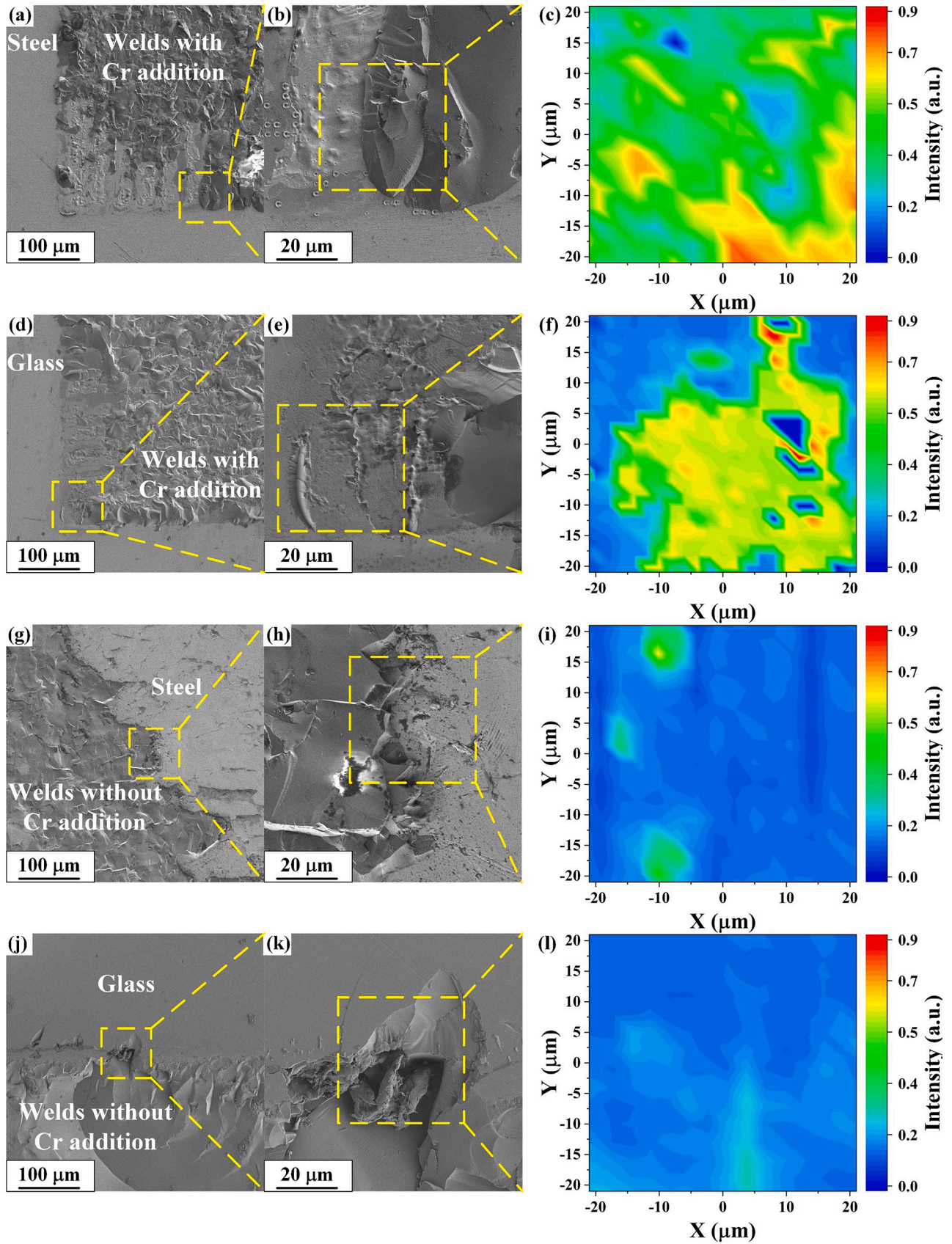


Fig. 10. SEM images of welds with Cr addition on (a, b) steel, (d, e) glass; photo of welds without Cr addition on (g, h) steel, (j, k) glass; Fe-O-Si bond Raman Mapping (c), (f), (i), and (l) of the areas in (b), (e), (h), and (k), respectively.

CRediT authorship contribution statement

Jingyu Huo: Conceptualization, Data curation, Formal analysis, Software, Writing – original draft. **Jinhui Yuan:** Investigation, Validation. **Qinrui Chen:** Investigation, Validation. **Minghuo Luo:** Investigation, Validation. **Junhao Lu:** Investigation, Validation. **Jiancheng Xu:** Investigation, Validation. **Chuangkai Li:** Investigation, Validation. **Aiping Luo:** Investigation, Validation. **Jiaming Li:** Methodology, Funding acquisition, Writing – review & editing. **Qingmao Zhang:** Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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